

Problem 11.8

The force of interest is given by

$$\delta_t = \frac{2}{k + 2t}.$$

It is known that:

1. A deposit of 75 at time $t = 0$ accumulates to X at time $t = 3$.
2. The present value at time $t = 3$ of a deposit of 150 at time $t = 5$ is also equal to X .

Calculate X .

Solution

1. Accumulation factor. The accumulation factor between times t_1 and t_2 is

$$\begin{aligned} a(t_1, t_2) &= \exp \left(\int_{t_1}^{t_2} \delta_t dt \right) = \exp \left(\int_{t_1}^{t_2} \frac{2}{k + 2t} dt \right) \\ &= \exp (\ln |k + 2t_2| - \ln |k + 2t_1|) = \frac{k + 2t_2}{k + 2t_1}, \end{aligned}$$

provided that $k + 2t$ is nonzero throughout the interval under consideration.

2. Value of both deposits at $t = 3$. From the first condition,

$$X = 75 a(0, 3) = 75 \frac{k + 6}{k}.$$

From the second condition, discount the deposit of 150 from $t = 5$ to $t = 3$:

$$X = \frac{150}{a(3, 5)} = 150 \frac{k + 6}{k + 10}.$$

3. Determining k and X . Equating the two expressions for X and canceling the nonzero factor $k + 6$,

$$\begin{aligned} 75 \frac{k + 6}{k} &= 150 \frac{k + 6}{k + 10} \\ k + 10 &= 2k \\ k &= 10. \end{aligned}$$

Therefore,

$$X = 75 \frac{10 + 6}{10} = 75 \frac{16}{10} = \boxed{120}.$$